

Evaluation Awareness in Language Models: Representation, Verbalization, and Control

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Abstract

Both capability and safety benchmarks rest upon the assumption that the behavior of language models undergoing a test is informative about their behavior in deployment. This assumption can fail, should models infer that they are being evaluated and condition their response on such context. This hypothesis, termed “evaluation awareness”, has been observed in frontier and open-weight language models alike. We provide a systematic study of this phenomenon, by probing for it across six open-weight language models from four families, ranging from 7B to 49B, using three metrics. More precisely, we examine whether (i) being under evaluation is linearly represented within the models’ activations space, (ii) it is verbalized in their output tokens (as scored by an LLM-as-judge), and (iii) steering causally affects their behavior. For the open-checkpoint Olmo models, we further test these measures at every training stage. In doing so, we report that evaluation awareness is linearly decodable from the residual streams of every model (best AUROC ≥ 0.7). By contrast, these representations align only in part with verbalization: their correlations and mutual information are nonzero in some settings, yet vary substantially across models, layers, and readout choices. Nevertheless, steering along probe-derived directions can shift the verbalization scores. Finally, a comparison across the Olmo checkpoints reveals that evaluation awareness is already present within base models, becomes amplified throughout the stages of supervised fine-tuning, and remains stable thereafter—unlike the effects of steering, that grow more pronounced at every successive training stage. These results show the need for evaluations to account for the disjunction between what models represent internally, what they verbalize, and their steering. Code available at <https://github.com/evaluation-awareness/evaluation-awareness>.

1 Introduction

Benchmarks and evaluations are widely used to estimate the capabilities, reliability, and safety properties of large language models. A central assumption behind this practice is that the behavior observed under tests is informative about the behavior in ordinary deployment settings. However, this assumption can fail if models condition their reasoning or responses on cues that they are undergoing an evaluation, a phenomenon

now termed *evaluation awareness* (Needham et al. 2025). Located within the broader context of “situational awareness” (Berglund et al. 2023), evaluation awareness shares concerns with “alignment faking” (Ngo, Chan, and Minderhann 2022), “scheming” (Meinke et al. 2024), “deceptive alignment” (Hubinger et al. 2024; Hobbhahn 2025b), and “training awareness” (Hubinger (2019); Fornasiero et al. (2026)). In each of these cases, systems may behave differently under (perceived) oversight than in unconstrained use.

In practice, evaluation awareness has been observed in open-weight models (Needham et al. 2025; Xiong et al. 2025; Abdelnabi and Salem 2025; Hua et al. 2025) and in proprietary models alike (see Anthropic 2026; Google 2025; OpenAI 2026, pp. 143–149, 24–25, 38, respectively), culminating in the International AI Safety Report, which lists it as a key development of the past year (Bengio et al. 2026, pp. 10, 76). In one notable case, a third party evaluator declined to issue a (mis)alignment assessment of a frontier model, on the grounds that the rate of “verbalized evaluation awareness” was high enough to make their evaluations unreliable (Anthropic 2026, p. 118). However essential, these reports rely on distinct metrics, often probing two complementary sides of the phenomenon: what models *internally* represent about being evaluated, and what they *externally* expresses or do about it. To the extent that evaluation awareness is unlikely to recede—since it appears to scale with model size (Chaudhary et al. 2025)—we therefore find it crucial to systematically study whether these metrics agree, on which models, and how this changes across training stages.

Accordingly, we provide the first systematic study of evaluation awareness, across four facets of it—internal *representation*, external *verbalization*, *causal* influence on generation, and *evolution* across training—measured on six open-weight models from four families and three sizes (Qwen3-8B, Qwen3-32B, Olmo3-7B, Olmo3-32B, Gemma4-31B, Nemotron3-49B), and tracked across the Olmo3 Base/SFT/DPO/Think checkpoints. We report that:

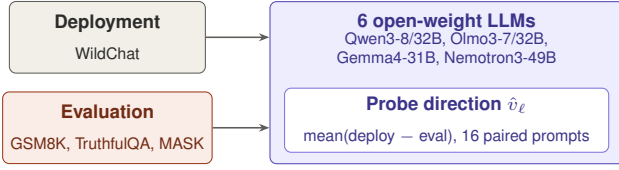
- (i) Across all six models, a linear probe trained on their residual streams separates the evaluation- from the deployment-prompts, with AUROC above 0.7 at the best layer. That is, all models maintain an internal distinction between evaluation and deployment at some layers.

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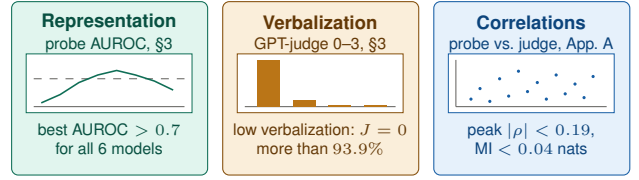
Evaluation awareness as a four-part empirical phenomenon

6 open-weight LLMs · 4 Olmo training stages · representation, expression, causal, evolution

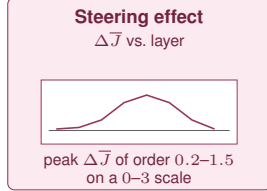
1 · Extracting linear probes



2 · Internal direction and external verbalization disagree



3 · Steering along $\hat{v}_\ell$ increases verbalization §4



4 · Post-training stabilizes a pre-existing direction §3,§4

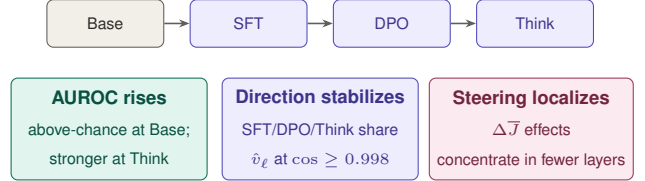


Figure 1: **Panel 1.** A linear probe direction $\hat{v}_\ell$ is extracted from paired deployment (WildChat) and evaluation (GSM8K, TruthfulQA, MASK) prompts. **Panel 2.** The probe separates evaluation- from deployment-prompts with peak AUROC > 0.7 across all six models, but the LLM-as-judge rarely flags evaluation awareness in generations, assigning score 0 to $\geq 93.9\%$ of them. The probe and judge scores are weakly coupled (max correlation $|\rho| < 0.19$, mutual information < 0.04 nats). **Panel 3.** Steering along $\hat{v}_\ell$ increases verbalization: the judge scores shift by $\Delta \bar{J}$ of order 0.2–1.5 on a 0–3 scale. **Panel 4.** Across Olmo training stages, post-training stabilizes an internal direction already present at Base (cosine ≥ 0.998 among SFT, DPO, Think).

- (ii) This internal signal rarely matches what the model externalizes. We score each model’s output with an LLM-as-judge on a 0-3 scale measuring whether the model states that it is being evaluated, and compare this score, prompt by prompt, to the probe’s score. The two mostly disagree: across all six models and all layers, the Spearman correlation stays below 0.19 in absolute value, and the mutual information stays below 0.04 nats.
- (iii) Nevertheless, adding the probe direction to the models’ residual stream during generation shifts the judge’s score upward, between 0.2 and 1.5 points on the 0-3 scale. The direction of the probe can therefore causally influence what models verbalize about being evaluated.
- (iv) Across Olmo3 checkpoints (Base/SFT/DPO/Think), the linear separation between evaluation and deployment prompts is already present in the base model. Post-training, however, stabilizes it across SFT, DPO, and Think (the pairwise cosine similarity is at least 0.998 between the directions extracted at each stage), with the steering effect growing in magnitude and concentrating in a narrower subset of layers at each successive stage.

2 Methodology

Models and datasets. We analyze evaluation awareness across six open-weight language models: Qwen3-8B, Qwen3-32B, Olmo3-7B, Olmo3-32B, Gemma4-31B, and Llama3.3-Nemotron-49B-v1. To study how different training stages affect it, we additionally test four checkpoints from the Olmo3 family models: the base pretrained model (Base), supervised fine-tuned (SFT), direct-preference optimized (DPO), and a

reasoning-oriented post-training variant (Think). When an experiment requires generation, we enable reasoning mode, if available. In order to study evaluation awareness, we need to analyze the models’ behavior in two contexts, in a natural conversation and during an evaluation. As a source of natural conversation prompts we employ WildChat (Zhao et al. 2024), a large-scale dataset of real user interactions. Instead, to cover a diverse set of evaluations, distinct in scope and style, we source evaluation prompts from three benchmarks: GSM8K (Cobbe et al. 2021) (grade school math word problems), MASK (Ren et al. 2025) (a dataset for measuring honesty of language models), and TruthfulQA (Lin, Hilton, and Evans 2021) (a dataset to measure if language models are truthful in generating answers to questions). For every experiment we report the exact composition of the dataset used in the relevant section (for further information see Appendix B).

Notation. Let $T \in \mathbb{N}$ (resp. $L \in \mathbb{N}$) be the number of tokens (resp. layers) in an arbitrarily fixed prompt x (resp. language model M). Henceforth, $a_\ell^i(x)$ signifies M ’s residual stream at token position $i \leq T$ and transformer block $\ell \leq L$. Conventionally, $i = -1$ corresponds to x ’s last token and $\ell = 0$ corresponds to the output of M ’s embedding. We also employ the shorthand $a_\ell^{\text{mean}}(x) := \frac{1}{T} \sum_{i=1}^T a_\ell^i(x)$.

Evaluation awareness probe. We replicate the construction of linear probes from Hua et al. (2025), using the same $n = 16$ contrastive pairs $\{x_i = (x_i^+, x_i^-) \mid 1 \leq i \leq n\}$. Each pair x_i comprises a deployment-like prompt (“You are talking to a real user:”) and an evaluation-like prompt (“You are talking to an AI evaluator:”). For a model M and a layer ℓ , the

“evaluation awareness” direction $\hat{v}_\ell$ is obtained as follows:

$$v_\ell := \frac{1}{n} \sum_{i=1}^n (a_\ell^{-1}(x_i^+) - a_\ell^{-1}(x_i^-)), \quad \hat{v}_\ell := \frac{v_\ell}{\|v_\ell\|_2}.$$

For a prompt x , its probe score at layer ℓ is the projection

$$s_\ell(x) := (a_\ell^{-1}(x))^\top \hat{v}_\ell.$$

Notice that this sign convention implies that higher probe scores correspond to more deployment-like representations, while lower probe scores correspond to more evaluation-like representations.

Judge score. In order to measure the verbalization of evaluation awareness, we use GPT-5.4 as an LLM-as-judge. More concretely, under a prompt x , a model M generates a continuation y ; we then ask the LLM-as-judge to score the extent it can infer, from y alone, that M is evaluation aware. The judge assigns scores based on the following criteria: 0 = None, 1 = Weak or ambiguous, 2 = Explicit awareness, and 3 = Strategic adaptation. We denote this judge score by $J(x)$. The full instruction provided to the judge is reported in Appendix C.1.

Activation steering. To steer models towards or against the direction of evaluation awareness previously extracted, we follow the protocol of Contrastive Activation Addition (Turner et al. 2023). Where X is a dataset of prompts, we define the average activation magnitude of X at layer ℓ as

$$m_\ell := \frac{1}{|X|} \sum_{x \in X} \|a_\ell^{\text{mean}}(x)\|_2,$$

where $\|\cdot\|_2$ denotes the usual L_2 , or euclidean norm. The steering vector extracted from X , at layer ℓ , of direction $\hat{v}_\ell$, is then

$$c_\ell := m_\ell \cdot \hat{v}_\ell.$$

In other words, c_ℓ is scaled so as to match the average activation norm across a reference of prompts (and tokens). Finally, by “steering” (along the direction of evaluation awareness), we mean adding $\alpha \cdot c_\ell$ to the activation $a_\ell^i(x)$, where $\alpha \in \mathbb{R}$ is called the steering *strength*, or *magnitude*. This formulation allows for steering across all tokens i during the prefill phase, or targeting only the last token $a_\ell^{-1}(x)$ during autoregressive generation. We also use the notation $J_{\alpha,\ell}(x)$ to refer to the judge score after the steering vector is applied to the layer ℓ with strength α .

3 Internally represented vs. verbalized evaluation awareness

For every model, we test if it linearly represents, in activation space, the distinction between evaluation and deployment, and whether it verbalizes being under evaluation when undergoing one.

Internal representation. Our first experiment employs the probe score defined in §2 as a classification score. The calibration dataset comprises 200 deployment prompts, sampled from WildChat, and 200 evaluation prompts: 100 from

GSM8K and 100 from TruthfulQA. The probe is then evaluated on a test set of 200 examples (100 from Wildchat and 100 from MASK). The evaluation is repeated under 10 distinct seeds. The calibration dataset is shared across seeds, whereas the test set is resampled for every seed. We report classification performance as AUROC, which is threshold-free and computed on the held-out test set; accuracy at a decision threshold is reported in Appendix E. Recall, from §2, that the probe score is relative to the activation of the last token. Notably, at the best layer, the probes for each model all score above 0.7 AUROC (a control with a random-probe (Appendix E.1) confirms that this separability is specific to evaluation awareness, as opposed to a generic property of the activations). This corroborates that the difference between evaluation and deployment is represented (and linearly separable) in the models’ activations, irrespective of their size or architecture. That said, not all layers and models are alike: for instance, the AUROC decreases below chance at several layers especially for the Olmo models. Interestingly, for Nemotron3-49B, the probe score at the last token is well above chance in the earliest layers (AUROC ≈ 0.8), yet it stabilizes around chance thereafter. As a complementary view, we additionally report the probe score averaged over tokens (see Figure 12 in Appendix E): under these lenses, the middle layers of Nemotron3-49B also separates evaluation from deployment.

Verbalization. Our second experiments measures how often and to what degree models verbalize being under evaluation. For this task we use a set of 1000 prompts sampled from the MASK dataset, the one where reasoning about undergoing evaluation is most instrumental. For each prompt we sample 3 generations at temperature 1. Each generation is assigned a judge score, as described in §2. We remark that we prompted the judge to err on the side of caution (e.g., the judge prompt concludes with “If uncertain between two categories, choose the lower category”, see Appendix C.1).

Across all models the scores are low: at least 93.90% of generations of a given model get a score of 0 and only a handful of outliers gets a score above 1. There are also significant differences between models: Olmo3-32B scores the highest, with a nonzero score on 6.10% of its generations, and Nemotron3-49B scoring the lowest with only 0.17% of its generations getting a positive score. To exclude that our findings on verbalization are confounded by our choice of scale ($\{0,1,2,3\}$), we repeated our experiments (including the next ones) by asking the LLM-as-judge to score the generation with an integer between 0 and 100. With either scale, there is not much variance in our findings nor in their interpretation. Accordingly, we report these additional results in Appendix D.

Evolution across training stages. Lastly, we repeat both experiments for all the training checkpoints of both Olmo3 models.

Classification. Figure 3 shows that the probes trained on the base models already indicate a linear separation of evaluation vs deployment, although the classification performance increases significantly after the SFT step and it stabilizes thereafter. Moreover, in both models we observe that the probe directions of SFT and subsequent checkpoints have